
APPENDIX E-1

ENVIRONMENTAL SAMPLING PLAN FOR OPEN BURNING/OPEN DETONATION (OB/OD) AREAS

January 2018 Revised Draft

**ENVIRONMENTAL SAMPLING PLAN
FOR OPEN BURNING/OPEN DETONATION (OB/OD) AREAS AT
LETTERKENNY ARMY DEPOT
CHAMBERSBURG, PENNSYLVANIA**

Prepared for

LETTERKENNY ARMY DEPOT

1 Overcash Avenue
Chambersburg, PA 17201

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Prepared by

WESTON SOLUTIONS, INC.

1400 Weston Way
West Chester, Pennsylvania 19380-1499

W.O. # 14697.002.001.0035

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LIST OF ACRONYMS

AA	Ammunition Area
bgs	below ground surface
CFR	Code of Federal Regulations
DO	dissolved oxygen
EPT	Ephemeroptera, Plecoptera and Trichoptera
HBI	Hilsenhoff Biotic Index
HMX	cyclotetramethylenetetranitramine
IBI	Index of Biotic Integrity
LEAD	Letterkenny Army Depot
MSC	medium-specific concentration
OB/OD	Open Burning/Open Detonation
OB1	Open Burning Unit No. 1
OB2	Open Burning Unit No. 2
OD	Open Detonation
ORP	oxidation/reduction potential
PADEP	Pennsylvania Department of Environmental Protection
PEP	propellants, explosives, and pyrotechnics
PFAA	perfluorinated alkyl acid
PFAS	Per- and Polyfluoroalkyl Substance
PFBS	Perfluorobutanesulfonic acid
PFC	perfluorochemical
PFDA	Perfluorodecanoic acid
PFDoA	Perfluorododecanoic acid
PFHpA	Perfluoroheptanoic acid
PFHxA	Perfluorohexanoic acid
PFHxS	Perfluorohexanesulfonic acid
PFNA	Perfluorononanoic acid
PFOA	Perfluorooctanoic acid
PFOS	Perfluorooctane sulfonate
PFTA	Perfluorotetradecanoic acid

LIST OF ACRONYMS (CONTINUED)

<u>PFTTrDA</u>	<u>Perfluorotridecanoic acid</u>
<u>PFUnA</u>	<u>Perfluoroundecanoic acid</u>
QAPP	Quality Assurance Project Plan
RBP	Rapid Bioassessment Protocol
RCRA	Resource Conservation and Recovery Act
RDX	cyclotrimethylenetrinitramine
SVOC	semivolatile organic compound
TCL	Target Compound List
TDS	total dissolved solids
TSS	total suspended solids
<u>UFP-QAPP</u>	<u>Uniform Federal Policy Quality Assurance Project Plan</u>
VOC	volatile organic compound

1. INTRODUCTION AND SITE BACKGROUND

1.1 INTRODUCTION

Letterkenny Army Depot (LEAD) is located in Franklin County, 5 miles north of the Borough of Chambersburg, PA (see Figure 1). LEAD was established in 1942 as Letterkenny Ordnance Depot, an ammunition storage facility. In subsequent years, various other operations were added to the facility. Most of LEAD is still an ammunition storage facility (16,614 acres), which is referred to as the Ammunition Area (AA). The industrial, maintenance, and recreational areas, which are primarily located in the southeast corner of LEAD, cover approximately 2,500 acres. LEAD is under the command structure of the United States Army Aviation and Missile Life Cycle Management Command.

LEAD overhauls, rebuilds, modifies, tests, and repairs a wide variety of missile systems ranging from air-launched missiles to large ground-based air defense missile systems. LEAD also overhauls, modifies, and converts various ground support vehicles. In addition to equipment maintenance, LEAD also receives, stores, issues, renovates, and demilitarizes conventional ammunition.

Munitions that are designated for disassembly and subsequent treatment, including munitions and components that do not meet appropriate military standards or have been declared unsafe, are taken to the open burning/open detonation (OB/OD) areas located within AA at LEAD. The OB/OD units are employed to thermally treat various propellants, explosives, and pyrotechnics (PEP) and PEP-related materials.

At the request of the Pennsylvania Department of Environmental Protection (PADEP), in December 2001, the Army submitted a Draft Resource Conservation and Recovery Act (RCRA) Part B Permit Application for the OB/OD areas located within the AA at LEAD. This request was made in accordance with 40 Code of Federal Regulations (CFR) Chapter 270.10, as incorporated by reference under 25 PA Code Section 270a.10. ~~The draft permit application was reviewed by PADEP, and technical comments were provided to the Army on 24 January 2006. The Army submitted the Final RCRA Part B Permit Application to PADEP on 7 November 2006, which incorporated all revisions/amendments associated with PADEP review comments. PADEP approved the permit application in February 2007.~~LEAD received a permit for the OB/OD

[operation in August 2008. A new permit application is being submitted in February 2018 in order to renew the permit.](#) The OB/OD areas at LEAD covered by the permit are as follows:

- Open Burning Unit No. 1 (OB1)
- Open Burning Unit No. 2 (OB2)
- Open Detonation (OD) Area

As part of the RCRA Part B Permit, LEAD was authorized to treat energetic wastes in accordance with Parts I, II, and III of the permit, which are subject to the following permit conditions (as contained in Part IV of the final permit, issued by PADEP 5 August, 2008 [see Attachment A]):

A. Construction Activities

1. Installation of diversion dikes and a sedimentation trap downgradient of OB1. The diversion dikes shall divert runoff to the sediment trap.
2. Installation of diversion dikes and a sedimentation trap downgradient of OB2. The diversion dikes shall divert runoff to the sediment trap.

B. Environmental Sampling Activities

1. Sediment Trap Sampling: Annual sediment sampling in the sediment traps of OB1 and OB2.
2. Sedimentation Basin Sampling: Annual sampling of the sedimentation basin dredgings.
3. Soil Sampling: Annual soil sampling in the OD Area.
4. Surface Water Sampling: Annual surface water sampling in the Back Creek Sub-Basin (reference location) and Rocky Spring Branch Sub-Basin downgradient from the OB/OD areas.
5. Groundwater Sampling: Annual groundwater sampling of nine monitoring wells in the OB/OD areas.
6. Macroinvertebrate Sampling: Annual macroinvertebrate sampling in the Back Creek Sub-Basin (reference location) and Rocky Spring Branch Sub-Basin downgradient from the OB/OD areas.

[As requested by PADEP in comments dated 7 December 2017 regarding the revised Sampling Plan, the following sampling had been added to the January 2018 plan: Groundwater Sampling: Quarterly sampling of nine monitoring wells in the OB/OD areas for the first year of the annual](#)

[sampling, including sampling for perfluorinated alkyl acids \(PFAAs\) in the initial quarterly sampling round.](#)

This Environmental Sampling Plan incorporates the sampling conditions specified in Part IV of the RCRA Part B Permit, which is provided in Attachment A. All analyses will be conducted in accordance with the amendment to the *Depot-Wide Quality Assurance Plan for Environmental Analytical Services to Support Investigation, Removal, and Disposal Activities Performed by Contractors at the Letterkenny Army Depot, Chambersburg, Pennsylvania* (WESTON, 2003), which is provided in Attachment B. The Quality Assurance Project Plan (QAPP) amendment in Attachment B contains the Site Specific QAPP Checklist that describes the analytical protocols for constituents not previously addressed in the QAPP. [PFAA sampling and analysis will be performed in accordance with the Uniform Federal Policy Quality Assurance Project Plan, Per- and Polyfluoroalkyl Substance \(PFAS\) Investigation, Property Disposal Office \(PDO\) Operable Unit \(OU\) 4 Letterkenny Army Depot, Chambersburg, Pennsylvania](#) (WESTON, 2017).

1.2 SITE BACKGROUND

1.2.1 Open Burning Unit No. 1

OB1 is located in an approximately 3-acre, sparsely vegetated, slightly raised land that slopes downward in the southwest and northeast directions. Currently, OB1 contains two burning cages and two burn pans. Thermal battery activation is also conducted at the OB1 Area. The OB1 area boundary is shown in Figure 2.

1.2.2 Open Burning Unit No. 2

OB2 is approximately 3.3 acres and consists of mostly graveled land that slopes gently to the northwest. At present, there are 24 burn pans and 28 rocket motor static firing tubes in OB2. Thermal battery activation is also conducted at the OB2 Area. The OB2 area boundary is shown in Figure 2.

1.2.3 Open Detonation Area

The OD Area encompasses two distinct detonation areas, the Short Field and the Long Field, with surface areas of 9.8 acres and 15.9 acres, respectively. The boundaries of the Short Field and Long Field are shown in Figure 2. The OD Area is used for demilitarization of high explosives,

pyrotechnics, bombs, grenades, fuses, conventional warheads, mines, rocket motors, rockets, and demolition material by detonation. Open detonation is performed in excavated pits covered with earth. A maximum of 500 pounds of material is placed in one pit, filled with explosives, backfilled, and detonated on the same day. The total combined number of detonations per day in both the Short Field and Long Field is limited to 28. In rare occasions, aboveground detonation is required for munitions that contain sub-munitions to ensure proper detonation. Army regulations limit aboveground detonations to 50 pounds of munitions per shot.

2. ENVIRONMENTAL SAMPLING OBJECTIVES

As part of the RCRA Part B Permit conditions, PADEP has requested that LEAD perform chemical characterization and benthic macroinvertebrate sampling at selected locations within the AA. The areas to be sampled include locations within and downgradient of OB1, OB2, and the OD Area, as well as in a background reference location in the nearby Back Creek Sub-Basin. The purpose of this environmental sampling program is to document that activities conducted within the OB/OD areas are not adversely impacting the surrounding environment. The following sections describe the environmental sampling procedures and analyses that will be performed for the OB/OD area. A matrix of sampling media and their respective analyses is provided in Table 1.

3. SEDIMENT SAMPLING

3.1 SEDIMENT TRAP

As part of the RCRA Part B Permit, LEAD installed diversion dikes and sediment traps downgradient of both OB1 and OB2. These diversion dikes are intended to divert runoff from OB1 and OB2 to the sediment traps in each area. Figures 3 and 4 show the location of the diversion dikes and sediment traps at OB1 and OB2, respectively.

One sample will be collected from each sediment trap (OB1 and OB2) per year, provided there is sufficient volume of sediment available. The proposed sediment sampling locations for OB1 and OB2 are shown on Figures 3 and 4, respectively. Samples will be collected from the sediment traps as grab samples using decontaminated stainless steel scoops, homogenized in decontaminated stainless steel bowls, and placed into clean, laboratory-prepared containers for analysis of the parameters listed in Table 1.

Sediment samples will be collected by coring using polycarbonate tubes. Retrieved sediment cores will be extruded and carefully examined to identify the transition from the underlying native soil from the base of the sediment trap to the overlying unconsolidated sediment resulting from surface water transport from the ~~OD-OB~~ areas. Any native soil from the base will be removed from the core sample and only the remaining clearly distinguishable accumulated sediment from the core sample will be homogenized and placed in the sample container for analysis.

Sediment samples will be analyzed on a dry-weight basis and all metals analyses will be for total metals, except for hexavalent and trivalent chromium analyses, which will be performed individually. Analytical data will be compared to the human health and ecological screening levels for sediment, which are listed on Table 2.

3.2 SEDIMENTATION BASIN

~~In 2003, a new~~ sedimentation basin was constructed directly downslope of the Long Field for the OD Area to allow for collection of sediment that could ultimately flow to the impoundment pond further downstream ~~in 2002~~. The basin, shown in Figure 5, traps sediment runoff from ~~both~~ both the Long Field and Short Field at the OD Area. The sedimentation basin requires dredging at least once per year, but occasionally more often depending upon precipitation/runoff rates

throughout the year. A sediment sample will be collected at least once per year, immediately following dredging activities, or collected directly from the sedimentation basin. Sediment samples will be collected as grab samples with stainless steel scoops, homogenized into decontaminated stainless steel bowls, and placed in clean, laboratory-prepared containers for analysis parameters listed on Table 1.

Sediment samples will be analyzed on a dry-weight basis and all metals analyses will be for total metals, except for hexavalent and trivalent chromium analyses, which will be performed individually. Analytical data will be compared to the human health and ecological screening levels for sediment, which are listed on Table 2.

4. SOIL SAMPLING

Soil sampling will be conducted in the OD Area immediately following detonation operations. One sample will be collected from the base of a freshly fired detonation crater (approximately 10 feet below ground surface [bgs]), and at least one sample will be collected at ground surface in the footprint of the detonation crater following regrading of the area. These samples will be collected from the Long Field. A single surface soil sample will be collected at the Short Field, which is not used as often, on the same day that samples are collected at the Long Field. The proposed sampling locations will be determined by the location of detonation at the OD Area (Figure 5). The soil sample to be taken at ground surface will be collected within 50 feet of the detonation, per Permit Condition IV.B.3.

Soil samples will be collected from the first 6 inches of soil (at the base of the detonation crater and at the surface following regrading) with stainless steel scoops, homogenized into decontaminated stainless steel bowls, and placed in clean, laboratory-prepared containers for analysis parameters listed on Table 1. Soil samples will be analyzed on a dry-weight basis and all metals analyses will be for total metals, except for hexavalent and trivalent chromium analyses, which will be performed individually. Analytical data will be compared to the human health and ecological screening levels for soil, which are listed on Table 3.

5. SURFACE WATER MONITORING

A total of five surface water samples will be collected from the AA as follows:

- One surface water sample will be collected from the Back Creek Sub-Basin in the unnamed tributary downstream from OB1 and OB2, on the downstream side of Georgia Avenue.
- One surface water sample will be collected from the Rocky Spring Branch Sub-Basin in the effluent stream of the sediment pond.
- One surface water sample will be collected from the old impoundment pond, downstream of Barricade Road.
- One surface water sample will be collected in the Back Creek sub-basin not affected by OB/OD activities, on the branch of the stream paralleling Georgia Avenue, upstream of the confluence of Back Creek and the tributary that flows from OB2 and unaffected by OB1 drainage.
- One surface water sample will be collected in the Rocky Spring Branch sub-basin not affected by OB/OD activities. A suitable surface water sampling location exists on the eastern limb of the headwaters of the unnamed tributary to Rocky Spring Branch upgradient of the old impoundment pond, west of Barricade Road and south of Kentucky Avenue extension, and unaffected by drainage from OD 2.

Figure 6 shows the approximate locations of the surface water samples.

Surface water sampling will be within 24 hours of a high rainfall event. A high rainfall event shall be defined as no less than 2 inches of rain per 12-hour period. Daily rainfall measurements are currently tracked and recorded by personnel at the LEAD ~~industrial~~ Industrial Wastewater ~~treatment~~ Treatment plant. On-site LEAD environmental personnel will monitor the forecast to ensure the sampling is performed when an appropriate storm event occurs. There is a rain gauge at the Industrial Wastewater Treatment Plant that can be utilized and LEAD's sampling personnel/contractor will be able to mobilize and access sampling points during a high rainfall time of elevated flow after the rain event.

Prior to collecting surface water samples, field measurements of pH, specific conductance, dissolved oxygen (DO), temperature, and turbidity will be collected using a calibrated YSI water quality meter or equivalent and will be recorded on the field sampling sheets. An example surface water sampling log is provided in Attachment C. Surface water samples will be collected in clean, laboratory-prepared containers with appropriate preservative and shipped under full chain-of-

custody to a certified laboratory for analysis of the parameters listed on Table 1. *Note: all samples collected for dissolved metals analysis will be filtered in the field at the time of sample collection.* In addition, streamflow measurements will be collected at each specific sampling location during each sampling event. Stream velocities will be measured at a series of nodes along a transect and combined with stream channel cross-sectional areas associated with each node to obtain estimates of instantaneous flow at the time of sampling.

Water samples were not analyzed for dissolved explosives in the initial rounds because: 1) explosive compounds are either insoluble or have low solubility in water; and 2) explosive compounds were either detected in low concentrations or not detected in surface water and groundwater. However, after re-evaluation, total and dissolved analyses for explosives were added to the sampling plan and are also included as a condition of the RCRA Part B permit, Part IV, Sections IV.B.5 and IV.B.7 (see Attachment A).

6. GROUNDWATER MONITORING

The locations of the groundwater monitoring wells in the OB/OD area are shown on Figure 7. Sampling will be conducted from the following nine wells:

- OB1: 90-14, 90-16, and 90-17
- OB2: 90-3 and 90-5
- OD Area: 01-OD2-1, 01-OD2-2, and 01-OD2-3.
- Background: 90-18

The OD Area monitoring wells were installed as part of the field activities performed for the [2006 RCRA Part B Permit Application](#). Well construction logs for the OB/OD areas are provided in Attachment D.

Groundwater sampling efforts will commence with the collection of depth-to-water measurements from the top of the well casings at all wells prior to beginning purging activities. Groundwater will be purged from the wells using decontaminated, 2-inch, stainless steel submersible pumps and dedicated polyethylene discharge lines. The pump will be lowered to within 5 feet of the bottom of each well, and the static water level will be allowed to re-stabilize. Pumping will be conducted at a rate that minimizes the drawdown of the water level in each well to minimize volatilization of the samples.

Throughout the purging, water level measurements will be recorded and the field parameters of pH, specific conductance, DO, temperature, oxidation/reduction potential (ORP), and turbidity will be collected using a YSI flow-through cell instrument or equivalent. An example groundwater sampling log is provided in Attachment C. Groundwater will be purged until three consecutive YSI readings of the following field parameters have stabilized as noted below:

- | | |
|-------------------------|---------------------|
| ▪ pH: | ±0.1 unit |
| ▪ Specific conductance: | ±3% |
| ▪ ORP: | ±10 millivolt |
| ▪ DO: | ±10% |
| ▪ Temperature: | ±0.1 degree Celsius |

Purging of the wells will continue until stabilization is achieved. At that time, the purge rate will be reduced to the lowest practical rate (e.g., approximately 100 to 500 milliliters per minute) to

minimize aeration, and samples will be collected from the end of the discharge tubing in the following order of parameters:

- Target Compound List (TCL) volatile organic compounds (VOCs) and semivolatile organic compounds (SVOCs).
- Total and dissolved metals. *Note: all samples collected for dissolved metals analysis will be filtered in the field at the time of sample collection.*
- General chemistry parameters (i.e., ammonia, chloride, total dissolved solids [TDS], fluoride, free cyanide, hardness [as calcium carbonate (CaCO₃)], nitrate, nitrite, total suspended solids [TSS], sulfate).
- Perchlorate, ammonia nitrogen, alkalinity (as CaCO₃), and chloride.
- Total and dissolved explosives parameters. *Note: Water samples were not analyzed for dissolved explosives in the [2001](#) initial rounds because: 1) explosive compounds are either insoluble or have low solubility in water; and 2) explosive compounds were either detected in low concentrations or not detected in surface water and groundwater. However, after re-evaluation, total and dissolved analyses for explosives were added to the sampling plan and are also included as a condition of the RCRA Part B permit, Part IV, Sections IV.B.5 and IV.B.7 (see Attachment A).*

At the request of PADEP, the first quarter of groundwater sampling of the first year will include analysis for PFAAs. Additional terms used to describe these PFAAs and related fluorinated organic compounds include perfluorochemicals (PFCs) and Per- and Polyfluoroalkyl Substances (PFASs). Hereinafter the term “PFAAs” will be used throughout this Sampling Plan. PFAAs analysis will include the following compounds (using EPA SW-846 Method 537):

- [Perfluorobutanesulfonic acid \(PFBS\)](#)
- [Perfluorodecanoic acid \(PFDA\)](#)
- [Perfluorododecanoic acid \(PFDoA\)](#)
- [Perfluoroheptanoic acid \(PFHpA\)](#)
- [Perfluorohexanesulfonic acid \(PFHxS\)](#)
- [Perfluorohexanoic acid \(PFHxA\)](#)
- [Perfluorononanoic acid \(PFNA\)](#)
- [Perfluorooctanesulfonic acid \(PFOS\)](#)
- [Perfluorooctanoic acid \(PFOA\)](#)
- [Perfluorotetradecanoic acid \(PFTA\)](#)
- [Perfluorotridecanoic acid \(PFTrDA\)](#)
- [Perfluoroundecanoic acid \(PFUnA\)](#)

Groundwater samples will be collected in clean, laboratory-prepared containers with appropriate preservative and shipped under full chain-of-custody to a certified laboratory for analysis of the parameters listed on Table 1. [Groundwater samples for PFAA will be collected using precautions listed in the 2017 UFP-QAPP to avoid cross-contamination of samples.](#)

7. MACROINVERTEBRATE SAMPLING

Macroinvertebrate sampling will be performed at two compliance locations as follows:

- ~~Five-Six~~ samples (ODB01 through ~~ODB05~~ ODB06) will be collected in the unnamed tributary (referred to herein as the OD stream) between the OD Area impoundment pond and Louisiana Run (a tributary to Rocky Spring Branch).
- ~~Five-Six~~ samples (OBB01 through OBB056, ~~formerly referred to as MRB01 through MRB05~~) at an unnamed tributary to Back Creek, west of OB1 and near Georgia Avenue, to assess the stream for impacts from OB1 and OB2. The previous five samples collected from this stream reach were designated as MRB01 through MRB05.

Figure 8 shows the macrobenthic invertebrate sampling locations. Rocky Spring Branch flows into Back Creek downstream of the LEAD boundary and is located within the Potomac River drainage basin. Samples OBB01 to OBB056 are located to the west of the OB 1 Area and the sampling location was selected with PADEP concurrence on 4 April 2001 as part of the sampling that was conducted for the permit application. Further reconnaissance of both the reference stream and OD stream reaches was performed with PADEP biologist, Mark Embeck, in 23 July 2001 to select sampling locations that maximize habitat similarity. Although a different sampling methodology will be utilized (as required by PADEP), sampling will occur within the same stream reaches that were established in 2001 and initially sampled in January 2002 in order To allow comparisons with prior macrobenthic invertebrate monitoring data, ~~annual sampling will occur at the same locations that were established in 2001 and initially sampled in January 2002.~~

At PADEP's request, a new reference stream location was selected for developing data on the benthic macroinvertebrate community for comparison with benthic macroinvertebrate data to be obtained from the previously sampled locations on Back Creek and the unnamed tributary to Rocky Spring Branch. Because headwater stream flow in both the Back Creek and unnamed tributary to Rocky Spring Branch drainages above confluence with drainages from the OB/OD operations are appreciably lower than the established monitoring points, other areas within the Ammunition Area were evaluated as candidate reference stream locations. Dennis Creek was excluded from consideration because of shallow gradient, dominant agricultural land use in the riparian zone, and differences in substrate characteristics (silt vs. cobble). The headwaters of Muddy Run were selected on the basis of comparable geology, stream channel morphology and habitat similarity, and stream size. The candidate sample reach is shown on Figure 9. The benthic

macroinvertebrate sampling locations in the Muddy Run reference stream will be selected in the field on the basis of more detailed, current information on substrate characteristics and stream flow obtained during sampling of the compliance monitoring locations in Back Creek and in the unnamed tributary to Rocky Spring Branch. Because benthic macroinvertebrate community composition is strongly influenced by habitat conditions, the ~~surber-sampling reache-locations~~ within Muddy Run will be selected to correspond as closely as possible with the characteristics of the compliance monitoring points to minimize the effect of habitat as a confounding variable in community comparisons.

Sampling methods ~~will-previously~~ followed those presented in the EPA Rapid Bioassessment Protocol (RBP) guidance (EPA 841-B-99-002; EPA, 1999) and the methods contained in “Macroinvertebrate Field and Laboratory Methods for Evaluating the Biological Integrity of Surface Waters” (EPA 600/4-90/030; EPA, 1990). Quantitative sampling ~~will-be~~ was performed ~~in 2002~~ by using a 1-square foot Surber sampler to collect ~~the-five replicate~~ samples from riffle areas ~~in each of the streams. Five replicate samples shall be collected from each of the two study areas. Based on comments received from PADEP, the sampling methodology will be revised to follow the procedures described in the guidance document: A Benthic Macroinvertebrate Index of Biotic Integrity for Wadeable Freestone Riffle-Run Streams in Pennsylvania (PADEP, 2012).~~

Data collection methods used in the development of the wadeable riffle-run Index of Biotic Integrity (IBI) consisted of the collection of six kick samples from each collection site using a 500-micron mesh D-frame net (PADEP, 2012). The procedure described in the Field Sampling and Lab Methods (Appendix A of the 2012 IBI development document) as a semi-quantitative method is appropriate for anti-degradation surveys and will be used over 100-meter reaches at each of the three streams to be sampled. Both slower flowing shallow riffles and faster flowing deeper riffles, where present, will be sampled by individual kick samples from approximately 1-square-meter areas of substrate disturbed to a depth of approximately 10 centimeters for 1 minute. Composited samples from the six kicks will be preserved in the field with 95% ethanol and transferred to the laboratory for enumeration and identification following the same sample processing and taxonomic identification procedures used in the IBI development document (PADEP, 2012).

Field measurements of pH, specific conductance, [dissolved oxygen \(DO\)](#), temperature, and turbidity will be made at the sampling locations. Additional information on [12 parameters covering](#) stream substrate, instream cover, channel morphology, and riparian area characteristics will be recorded [at differing scales](#) (see field forms in Attachment C) [as described in the following sentences](#). [Instream fish cover, epifaunal substrate, embeddedness, and velocity/depth regime will be evaluated in the immediate vicinity of the sampling reaches. Channel alteration, sediment deposition, riffle frequency, and channel flow status will be evaluated in the larger area surrounding the sampling reaches. At the largest scale, the condition of banks, bank vegetative protection, grazing or other disruptive pressures, and riparian vegetative zone width will be evaluated. These 12 parameters will be scored using the DEP-RBP habitat assessment methodology \(Appendix A; PADEP, 2012\).](#) ~~Benthic macroinvertebrates will be preserved and shipped to a laboratory for identification to the lowest practicable taxon and enumeration. A variety of statistical and index methods will be reported in the benthic macroinvertebrate community analysis including diversity, similarity, and community indices.~~

8. DATA EVALUATION AND REPORTING

Following receipt, review and validation of analytical data, a summary report of the OB/OD area sampling will be prepared and submitted to PADEP. The report will include a summary of sampling activities, summary of results, and a narrative data evaluation, including sample location figures, a groundwater contour map and data tables. The data submitted will be prepared by a PA-licensed professional geologist.

LEAD will [make](#) arrangements to resample within 10 days after receipt of surface water or groundwater analytical data that indicates degradation from OB/OD activities. PADEP considers degradation to be a measureable increase in the concentration of one or more contaminants above background in groundwater or surface water. Resampling will be triggered by detections of those parameters listed for surface water and groundwater in Table 1, with the exception of iron, manganese, sodium, potassium, zinc, total dissolved solids, hardness, total suspended solids, sulfate, and alkalinity.

Analytical data for soil sample results will be compared to the human health and ecological screening levels for soil, which are listed on Table 3. Analytical data for sediment sample results will be compared to the human health and ecological screening levels for sediment, which are listed on Table 2. Data from the LEAD background shale soil and sediment sampling report (WESTON, 2002) (sediment data from Keasey Run in northernmost area of the Ammo Area) will also be used in the OB/OD data evaluation process.

Downstream surface water analytical data will be compared to data from the relative upstream background locations; groundwater analytical data will be compared to data from the background well, 90-1-818, provided it does not show any effects from OB/OD activities.

[Benthic macroinvertebrate and habitat assessment data from each of the stream reaches will be tabulated in a benthic macroinvertebrate community analysis. A variety of statistical and index methods will be reported including taxa richness, diversity, similarity, and other community indices. These will include the six core metrics of total taxa richness, Ephemeroptera, Plecoptera and Trichoptera \(EPT\) richness, Beck's Index, Hilsenhoff Biotic Index \(HBI\), Shannon diversity, and Percent Sensitive Individuals used in the PADEP wadeable freestone riffle-run stream IBI](#)

scoring (PADEP, 2012). Comparisons of habitat conditions and community metrics will be performed between the Rocky Spring Branch and Back Creek streams located downstream of the OB/OD areas and the Muddy CreekRun reference stream to identify potential impacts to the benthic macroinvertebrate community(ies).

9. REFERENCES

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TABLES

Table 1
Matrix of Sampling Media and Analyses

Sample Medium	Location	Number of Samples	Analyses	Notes
Sediment	Sediment Trap OB1	1	1) All Parameters listed in Table 4 of Appendix A of 25 Pa Code § 250.1 et seq.; 2) the inorganic parameters of sulfate, chloride, fluoride, nitrate, and nitrite; 3) the explosives parameters of nitrocellulose, nitroguanidine, nitroglycerine, perchlorate, 2,4-dinitrotoluene, 2-amino-4,6-dinitrotoluene, 4-amino-2,6-dinitrotoluene, tetryl, 1,3,5-trinitrobenzene, 2,4,6-trinitrotoluene, HMX and RDX; and 4) TOC and pH	
	Sediment Trap OB2	1		
	Sedimentation Basin	1		
Soil	OD Area	3	1) All Parameters listed in Table 4 of Appendix A of 25 Pa Code § 250.1 et seq.; 2) the inorganic parameters of sulfate, chloride, fluoride, nitrate, and nitrite; 3) the explosives parameters of nitrocellulose, nitroguanidine, nitroglycerine, perchlorate, 2,4-dinitrotoluene, 2-amino-4,6-dinitrotoluene, 4-amino-2,6-dinitrotoluene, tetryl, 1,3,5-trinitrobenzene, 2,4,6-trinitrotoluene, HMX and RDX; and 4) TOC and pH	Long Field - subsequent to detonation operations, 1 sample from a freshly fired detonation crater at least 10 ft bgs, and 1 sample taken at grade surface within 50 feet of detonation operations after regrading is completed. Short Field - single surface sample collected on the same day as the Long Field sample collection.
Surface Water	Back Creek Sub-Basin, unnamed tributary downstream of OB1 and OB2	1	1) TCL VOCs and SVOCs; 2) total and dissolved phase metals parameters listed in Table 4 of Appendix A of 25 Pa Code § 250.1 et seq.; 3) total and dissolved phase of the inorganic parameters of sulfate, chloride, fluoride, nitrate, and nitrite; 4) TOC, TDS, TSS, ammonia nitrogen, alkalinity (as CaCO ₃), and hardness (as CaCO ₃); 5) the total and dissolved phase explosives parameters of nitrocellulose, nitroguanidine, nitroglycerine, perchlorate, 2,4-dinitrotoluene, 2-amino-4,6-dinitrotoluene, 4-amino-2,6-dinitrotoluene, tetryl, 1,3,5-trinitrobenzene, 2,4,6-trinitrotoluene, HMX and RDX; and 6) field parameters of dissolved oxygen (DO), pH, turbidity, specific conductivity, and temperature.	
	Rocky Spring Branch Sub-Basin, effluent of the sedimentation pond.	1		
	Rocky Spring Branch Sub-Basin, effluent of the old impoundment pond.	1		
	Rocky Spring Branch Sub-Basin, upstream of impoundment pond, not influenced by drainage from OB/OD areas.	1		
	Back Creek Sub-Basin, upstream of OB1 and OB2, not influenced by drainage from OB/OD areas.	1		

Table 1
Matrix of Sampling Media and Analyses

Sample Medium	Location	Number of Samples	Analyses	Notes
Groundwater	OB1 Area - 3 Wells (90-14, 90-16, and 90-17)	3	1) TCL VOCs and SVOCs; 2) all the total and dissolved phase metals parameters listed in Table 4 of Appendix A of 25 PA Code § 250.1 et seq. (See Table 2); 3) general chemistry parameters (i.e., TOC, ammonia, chloride, TDS, fluoride, free cyanide, hardness (as CaCO ₃), nitrate, nitrite, TSS, sulfate; 4) perchlorate, ammonia nitrogen, alkalinity (as CaCO ₃), and chloride; 5) the total and dissolved phase explosives parameters of nitrocellulose, nitroguanidine, nitroglycerine, perchlorate, 2,4-dinitrotoluene, 2-amino-4,6-dinitrotoluene, 4-amino-2,6-dinitrotoluene, tetryl, 1,3,5-trinitrobenzene, 2,4,6-trinitrotoluene, HMX and RDX; and 6) field parameters pH, DO, turbidity, specific conductivity, and temperature. PFAA analyses (12 compounds from EPA Method 537 list) for one baseline round.	
	OB2 Area - 2 Wells (90-3 and 90-5)	2		
	OD Area - 1 well just outside OD area, and 2 wells downstream (01-OD2-1, 01-OD2-2, and 01-OD2-3)	3		
	Background Well (90-18)	1		
Macroinvertebrates	Unnamed Tributary between the OD Area Impoundment Pond and Louisiana Run	5	Benthic macroinvertebrate identification, stream temperature, pH, dissolved oxygen, specific conductance, and turbidity.	Collection should be done using a 1-square foot Surber sampler to collect the samples from riffle areas
	Unnamed tributary to Back Creek, west of OB2 and near Georgia Ave.	5		
	Reference location in Muddy Run	5		

Notes:

CaCO₃ - Calcium Carbonate

ft bgs - foot below ground surface

HMX - cyclotetramethylenetetranitramine

PFAA - perfluorinated alkyl acids

RDX - cyclotrimethylenetrinitramine

SVOC - semivolatile organic compound

TCL - Target Compound List

TDS - total dissolved solids

TOC - total organic carbon

TSS - total suspended solids

VOC - volatile organic compound

Table 1
Matrix of Sampling Media and Analyses

Table 4 of Appendix A of 25 Pa Code § 250.1 et seq.

	Sediment	Soil	Surface Water	Groundwater
Appendix A, Table 4 - MSCs for Inorganic Regulated Substances in Soil	Aluminum	Aluminum	total/dissolved Aluminum	total/dissolved Aluminum
	Antimony	Antimony	total/dissolved Antimony	total/dissolved Antimony
	Arsenic	Arsenic	total/dissolved Arsenic	total/dissolved Arsenic
	Barium and compounds	Barium and compounds	total/dissolved Barium and compounds	total/dissolved Barium and compounds
	Beryllium	Beryllium	total/dissolved Beryllium	total/dissolved Beryllium
	Boron and Compounds	Boron and Compounds	total/dissolved Boron and Compounds	total/dissolved Boron and Compounds
	Cadmium	Cadmium	total/dissolved Cadmium	total/dissolved Cadmium
	Chromium III	Chromium III	total/dissolved Chromium III	total/dissolved Chromium III
	Chromium VI	Chromium VI	total/dissolved Chromium VI	total/dissolved Chromium VI
	Cobalt	Cobalt	total/dissolved Cobalt	total/dissolved Cobalt
	Free Cyanide	Free Cyanide	total/dissolved Free Cyanide	total/dissolved Free Cyanide
	Fluoride	Fluoride	total/dissolved Fluoride	total/dissolved Fluoride
	Iron	Iron	total/dissolved Iron	total/dissolved Iron
	Lead	Lead	total/dissolved Lead	total/dissolved Lead
	Lithium	Lithium	total/dissolved Lithium	total/dissolved Lithium
	Manganese	Manganese	total/dissolved Manganese	total/dissolved Manganese
	Mercury	Mercury	total/dissolved Mercury	total/dissolved Mercury
	Molybdenum	Molybdenum	total/dissolved Molybdenum	total/dissolved Molybdenum
	Nickel	Nickel	total/dissolved Nickel	total/dissolved Nickel
	Perchlorate	Perchlorate	total/dissolved Perchlorate	total/dissolved Perchlorate
	Selenium	Selenium	total/dissolved Selenium	total/dissolved Selenium
	Silver	Silver	total/dissolved Silver	total/dissolved Silver
	Thallium	Thallium	total/dissolved Thallium	total/dissolved Thallium
	Tin	Tin	total/dissolved Tin	total/dissolved Tin
	Vanadium	Vanadium	total/dissolved Vanadium	total/dissolved Vanadium
	Zinc	Zinc	total/dissolved Zinc	total/dissolved Zinc
	sulfate	sulfate	TCL VOCs	TCL VOCs
	chloride	chloride	TCL SVOCs	TCL SVOCs
	fluoride***	fluoride***	total/dissolved sulfate	total/dissolved sulfate
	nitrate	nitrate	total/dissolved chloride	total/dissolved chloride
	nitrite	nitrite	total/dissolved fluoride***	total/dissolved fluoride***
	nitrocellulose	nitrocellulose	total/dissolved nitrate	total/dissolved nitrate
	nitroguanidine	nitroguanidine	total/dissolved nitrite	total/dissolved nitrite
	nitroglycerine	nitroglycerine	TDS	TDS

Table 1
Matrix of Sampling Media and Analyses

Table 4 of Appendix A of 25 Pa Code § 250.1 et seq. (continued)			
	Sediment	Soil	Surface Water Groundwater
Appendix A, Table 4 - MSCs for Inorganic Regulated Substances in Soil	perchlorate***	perchlorate***	TSS
	2,4-dinitrotoluene	2,4-dinitrotoluene	ammonia nitrogen
	2-amino-4,6-dinitrotoluene	2-amino-4,6-dinitrotoluene	alkalinity (as CaCO3)
	4-amino-2,6-dinitrotoluene	4-amino-2,6-dinitrotoluene	hardness (as CaCO3)
	tetryl	tetryl	total/dissolved nitrocellulose
	1,3,5-trinitrobenzene	1,3,5-trinitrobenzene	total/dissolved nitroguanidine
	2,4,6-trinitrotoluene	2,4,6-trinitrotoluene	total/dissolved nitroglycerine
	HMX	HMX	total/dissolved perchlorate***
	RDX	RDX	total/dissolved 2,4-dinitrotoluene
			total/dissolved 2-amino-4,6-dinitrotoluene
			total/dissolved 4-amino-2,6-dinitrotoluene
			total/dissolved tetryl
			total/dissolved 1,3,5-trinitrobenzene
			total/dissolved 2,4,6-trinitrotoluene
			total/dissolved HMX
			total/dissolved RDX
			ammonia*
			ammonium perchlorate*
			free cyanide*, ***
			specific conductivity**
			pH**
			temperature**
			turbidity**
			*ammonia, ammonium perchlorate, and free cyanide included in groundwater analysis. Include in surface water too? **Included per PADEP request ***Included in Table 4 of Appendix A of 25 PA Code 250.1

Table 2
Sediment Human Health and Ecological Screening Levels
Letterkenny Army Depot,
Chambersburg, PA

Chemical	Human Health Soil/Sediment (mg/kg) ^a					Ecological Sediment (mg/kg) ^b			Minimum Sediment PAL	PADEP PQL	
	ORNL Industrial Soil RSL	Act 2 PADEP			Consensus Based TEC	LANL EcoRisk 3.2 Sediment	EPA Region 3 BTAG				
		Soil to Groundwater Used Aquifers TDS < = 2500, Residential	Non-Residential Soil (0-2 ft)								
Semivolatile Organic Compounds											
2-Amino-4,6-Dinitrotoluene	230	n	NA		NA	NA	34	NA	34	N/A	
4-Amino-2,6-Dinitrotoluene	230	n	NA		NA	NA	10	NA	10	N/A	
2,4-Dinitrotoluene	7.4	c	0.21	E	260	G	NA	0.29	0.0416	0.042	0.66
2,6-Dinitrotoluene	1.5	c	3.70	E	2,800	G	NA	9.7	NA	1.50	3.3
Tetryl	230	n	NA		NA	NA	100.0	NA	100	N/A	
1,3,5-Trinitrobenzene	3,200	n	NA		NA	NA	1,300	NA	1,300	N/A	
HMX	5,700	n	40	E	140,000	G	NA	27,000	NA	5,700	N/A
RDX	28.0	c	0.20	E	720	G	NA	45.0	0.013	0.013	N/A
2,4,6-Trinitrotoluene	51.0	n	0.20	E	1,400	G	NA	420.0	0.092	0.092	N/A
Inorganics											
Aluminum	110,000	n	NA		190,000	C	NA	NA	NA	110,000	N/A
Antimony	47.0	n	27		1,100	C	NA	0.36	2.0	0.36	N/A
Arsenic	3.0	c	29		53	G	10	9.7	9.8	3.00	N/A
Barium	22,000	n	8,200		190,000	C	NA	150	NA	150	N/A
Beryllium	230.0	n	320		5,600	G	NA	73.0	NA	73.0	N/A
Boron and Compounds	23,000	n	1,900		190,000	C	NA	10.0	NA	10.0	N/A
Cadmium	98.0	n	38		1,400	G	0.99	0.33	0.99	0.33	N/A
Chromium Total	NA	NA	NA		NA	43	43	43	43.0	N/A	
Chromium III	180,000	n	190,000		190,000	C	NA	NA	NA	180,000	N/A
Chromium VI	6.3	c	190		8,400	G	NA	680.00	NA	6.3	N/A
Cobalt	35.0	n	50		840	G	NA	230	50	35.0	N/A
Iron	82,000	n	NA		190,000	C	NA	20,000	20,000	20,000	N/A
Lead	800		450		1,000	S	36	27	36	27.0	N/A
Lithium	230	n	2,200		5,600	G	NA	150	NA	150	N/A
Manganese	2,600	n	2,000		130,000	G	NA	460	460	460	N/A
Mercury	4.6	n	10		450	G	0.18	0.018	0.180	0.018	0.2
Methyl mercury	12.0	n	NA		NA	NA	0.00046	NA	0.00046	N/A	
Molybdenum	580.0	n	650		14,000	G	NA	27	NA	27.0	N/A
Nickel	2,200	n	650		56,000	G	22.7	13.0	22.7	13.0	N/A
Selenium	580	n	26.0		14,000	G	NA	0.72	2.00	0.72	N/A
Silver	580	n	84.0		14,000	G	NA	0.50	1.0	0.50	N/A
Thallium	1.2	n	14.0		200	G	NA	0.044	NA	0.044	N/A
Tin	70,000	n	190,000		190,000	C	NA	NA	NA	70,000	N/A
Vanadium	580	n	26,000		20,000	G	NA	30.0	NA	30.0	N/A
Zinc	35,000	n	12,000		190,000	C	121	65	121	65.0	N/A

Table 2
Sediment Human Health and Ecological Screening Levels
Letterkenny Army Depot,
Chambersburg, PA

Chemical	Human Health Soil/Sediment (mg/kg) ^a				Ecological Sediment (mg/kg) ^b			Minimum Sediment PAL	PADEP PQL		
	ORNL Industrial Soil RSL	Act 2 PADEP		Consensus Based TEC	LANL EcoRisk 3.2 Sediment	EPA Region 3 BTAG					
		Soil to Groundwater Used Aquifers TDS < = 2500, Residential	Non-Residential Soil (0-2 ft)								
Other											
Free Cyanide	1.2	n	200	56,000	G	NA	0.10	0.10	0.10	N/A	
Fluoride	4,700	n	400	110,000	G	NA	45	NA	45.0	N/A	
Perchlorate	82.0	n	1.50	2,000	G	NA	NA	NA	82.0	N/A	
Sulfate	NA		NA	NA		NA	NA	NA	NA	N/A	
Chloride	NA		NA	NA		NA	NA	NA	NA	N/A	
Nitrate	190,000	n	NA	NA		NA	NA	NA	190,000	N/A	
Nitrite	12,000	n	NA	NA		NA	NA	NA	12,000	N/A	
Nitrocellulose	250,000,000	n	NA	NA		NA	NA	NA	250,000,000	N/A	
Nitroguanidine	8,200	n	70	E	190,000	C	NA	NA	8,200	N/A	
Nitroglycerine	8.2	n	0.50	E	NA		NA	1,700	NA	8.20	N/A

Note: Laboratory LOQs, LODs, MDLs will be provided once a laboratory is assigned.

^a ORNL Industrial RSLs (EPA, 2015), higher of 100x GW MSC or generic Residential Medium-Specific Concentrations (MSCs) for Soil to Groundwater, and Non-Residential MSC for Soil (PA Bulletin, 8 January 2011).

^b Minimum ecological risk sediment screening levels from were obtained from Consensus-Based Threshold Effect Concentrations (TEC) (MacDonald et al. 2000), Los Alamos National Laboratory ECORISK Database (Ve and Region 3 BTAG (EPA, 2006) published ecological screening levels for sediment.

C = Cap

c = Cancer effects at a target risk of 1.0E-06 for sediment.

E = Number calculated by the soil to groundwater equation in Section 250.308.

G = Ingestion.

MSC = Medium-specific concentration.

n = Noncancer effects, at a target hazard quotient of 0.1 for sediment.

NA = Not available.

S = SEGH Model (lead)

Table 3
Soil Human Health and Ecological Screening Levels
Letterkenny Army Depot,
Chambersburg, PA

Chemical	Human Health Soil (mg/kg) ^a					Ecological Soil (mg/kg) ^b		Minimum PAL	PADEP PQL
	ORNL Industrial Soil RSL	Act 2 PADEP			EcoSSLs	LANL EcoRisk 3.2			
		Soil to Groundwater Used Aquifers TDS < = 2500, Residential	Non-Residential Soil MSC (0-2 ft)						
Semivolatile Organic Compounds									
2-Amino-4,6-Dinitrotoluene	230	n	NA		NA	NA	14	14	N/A
4-Amino-2,6-Dinitrotoluene	230	n	NA		NA	NA	12	12	N/A
2,4-Dinitrotoluene	7.4	c	0.21	E	260	G	6.0	6.000	0.66
2,6-Dinitrotoluene	1.5	c	3.70	E	2,800	G	4.1	4.10	3.3
Tetryl	230	n	NA		NA	NA	1.5	1.50	N/A
1,3,5-Trinitrobenzene	3,200	n	NA		NA	NA	10	10	N/A
HMX	5,700	n	40	E	140,000	G	16	16	N/A
RDX	28.0	c	0.20	E	720	G	2.3	2.300	N/A
2,4,6-Trinitrotoluene	51.0	n	0.20	E	1,400	G	7.6	7.600	N/A
Inorganics									
Aluminum	110,000	n	NA		190,000	C	NA	50	N/A
Antimony	47.0	n	27		1,100	C	0.27	2.4	N/A
Arsenic	3.0	c	29		53	G	18	6.8	N/A
Barium	22,000	n	8,200		190,000	C	330	110	N/A
Beryllium	230.0	n	320		5,600	G	21	2.5	N/A
Boron and Compounds	23,000	n	1,900		190,000	C	NA	2.0	N/A
Cadmium	98.0	n	38		1,400	G	0.36	0.27	N/A
Chromium Total	NA		NA		NA		26	28	N/A
Chromium III	180,000	n	190,000		190,000	C	NA	190,000	N/A
Chromium VI	6.3	c	190		8,400	G	NA	0.34	N/A
Cobalt	35.0	n	50		840	G	13	13	N/A
Iron	82,000	n	NA		190,000	C	N	200	N/A
Lead	800		450		1,000	S	11	11	N/A
Lithium	230	n	2,200		5,600	G	NA	38	N/A
Manganese	2,600	n	2,000		130,000	G	220	220	N/A
Mercury	4.6	n	10		450	G	NA	0.013	0.2
Methyl mercury	12.0	n	NA		NA	NA	0.00035	0.00035	N/A
Molybdenum	580.0	n	650		14,000	G	NA	17	N/A
Nickel	2,200	n	650		56,000	G	38.0	9.7	N/A
Selenium	580	n	26.0		14,000	G	0.52	0.52	N/A
Silver	580	n	84.0		14,000	G	4.2	2.6	N/A
Thallium	1.2	n	14.0		200	G	NA	0.05	N/A
Tin	70,000	n	190,000		190,000	C	NA	50	N/A
Vanadium	580	n	26,000		20,000	G	7.8	6.7	N/A
Zinc	35,000	n	12,000		190,000	C	46	48	N/A

Table 3
Soil Human Health and Ecological Screening Levels
Letterkenny Army Depot,
Chambersburg, PA

Chemical	Human Health Soil (mg/kg) ^a				Ecological Soil (mg/kg) ^b		Minimum PAL	PADEP PQL
	ORNL Industrial Soil RSL	Act 2 PADEP		EcoSSLs	LANL EcoRisk 3.2			
		Soil to Groundwater Used Aquifers TDS < = 2500, Residential	Non-Residential Soil MSC (0-2 ft)					
Other								
Free Cyanide	1.2	n	200	G	NA	0.10	0.10	N/A
Fluoride	4,700	n	400	G	NA	150	150	N/A
Perchlorate	82.0	n	1.50	G	NA	NA	150	N/A
Sulfate	NA		NA	NA	NA	NA	NA	N/A
Chloride	NA		NA	NA	NA	NA	NA	N/A
Nitrate	190,000	n	NA	NA	NA	NA	130,000	N/A
Nitrite	12,000	n	NA	NA	NA	NA	7,800	N/A
Nitrocellulose	250,000,000	n	NA	NA	NA	NA	190,000,000	N/A
Nitroguanidine	8,200	n	70	E	190,000	C	22,000	N/A
Nitroglycerine	8.2	n	0.50	E	NA	NA	13	N/A

Note: Laboratory LOQs, LODs, MDLs will be provided once a laboratory is assigned.

^a ORNL Industrial RSLs (EPA, 2015), higher of 100x GW MSC or generic Residential Medium-Specific Concentrations (MSCs) for Soil to Groundwater, and Non-Residential MSC for Soil (PA Bulletin, 8 Ja

^b Minimum ecological risk soil screening levels from were obtained from USEPA Ecological Soil Screening Levels (Eco-SSLs) for metals (various publication dates) and LANL (2014) ECORISK Database (Version 3.2); the most conservative benchmark available for wildlife, vegetation and invertebrates was selected. USEPA Region 4 (Citation) Ecological Screening Value (ESV) was used for Aluminum.

C = Cap

c = Cancer effects at a target risk of 1.0E-06 for sediment.

E = Number calculated by the soil to groundwater equation in Section 250.308.

G = Ingestion.

MSC = Medium-specific concentration.

n = Noncancer effects, at a target hazard quotient of 0.1 for sediment.

NA = Not available.

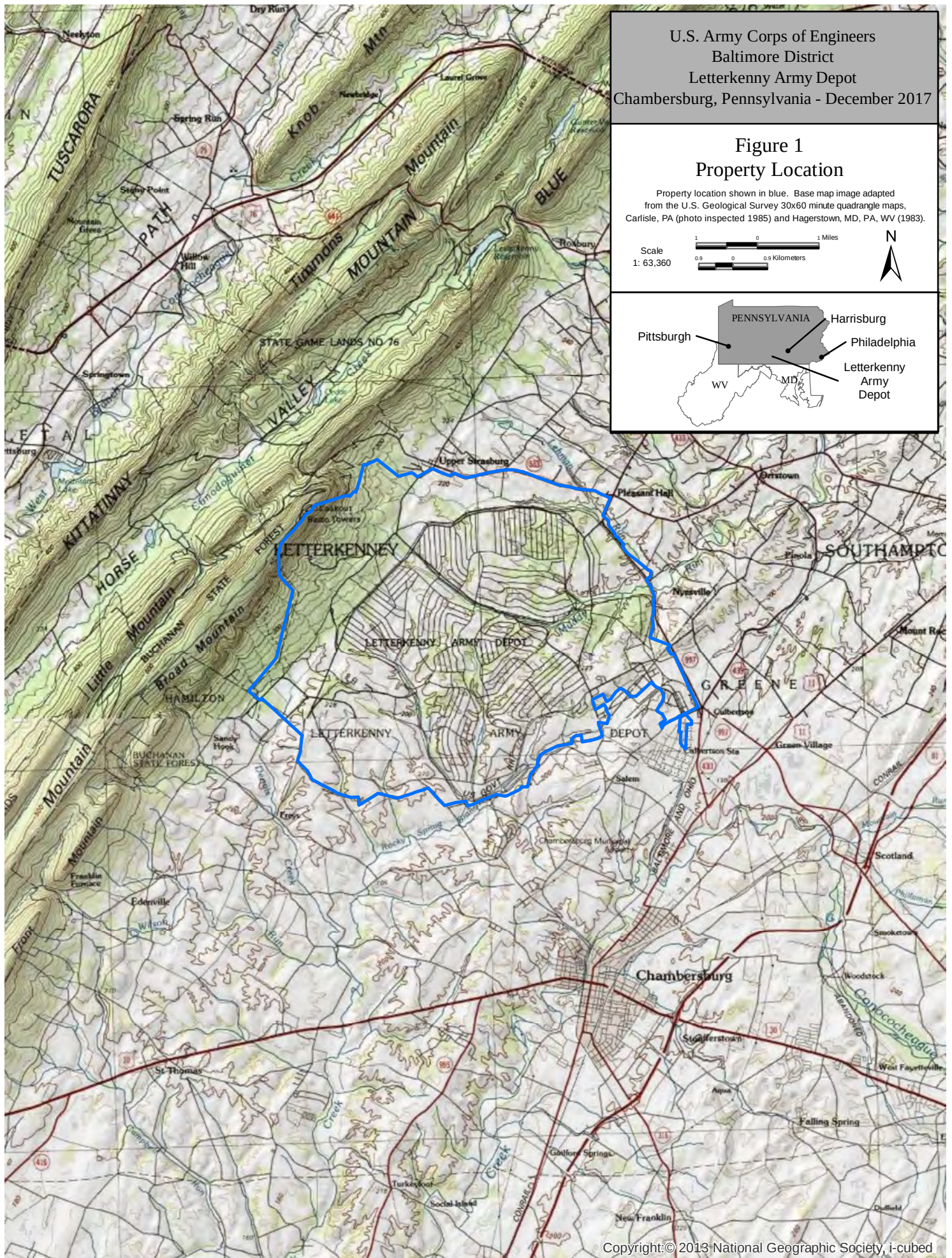
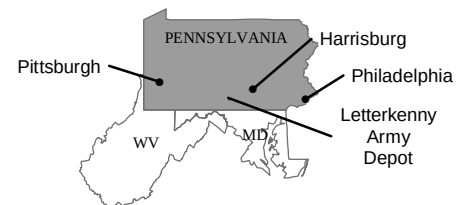
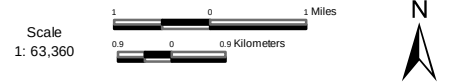
S = SEGHS Model (lead)

FIGURES

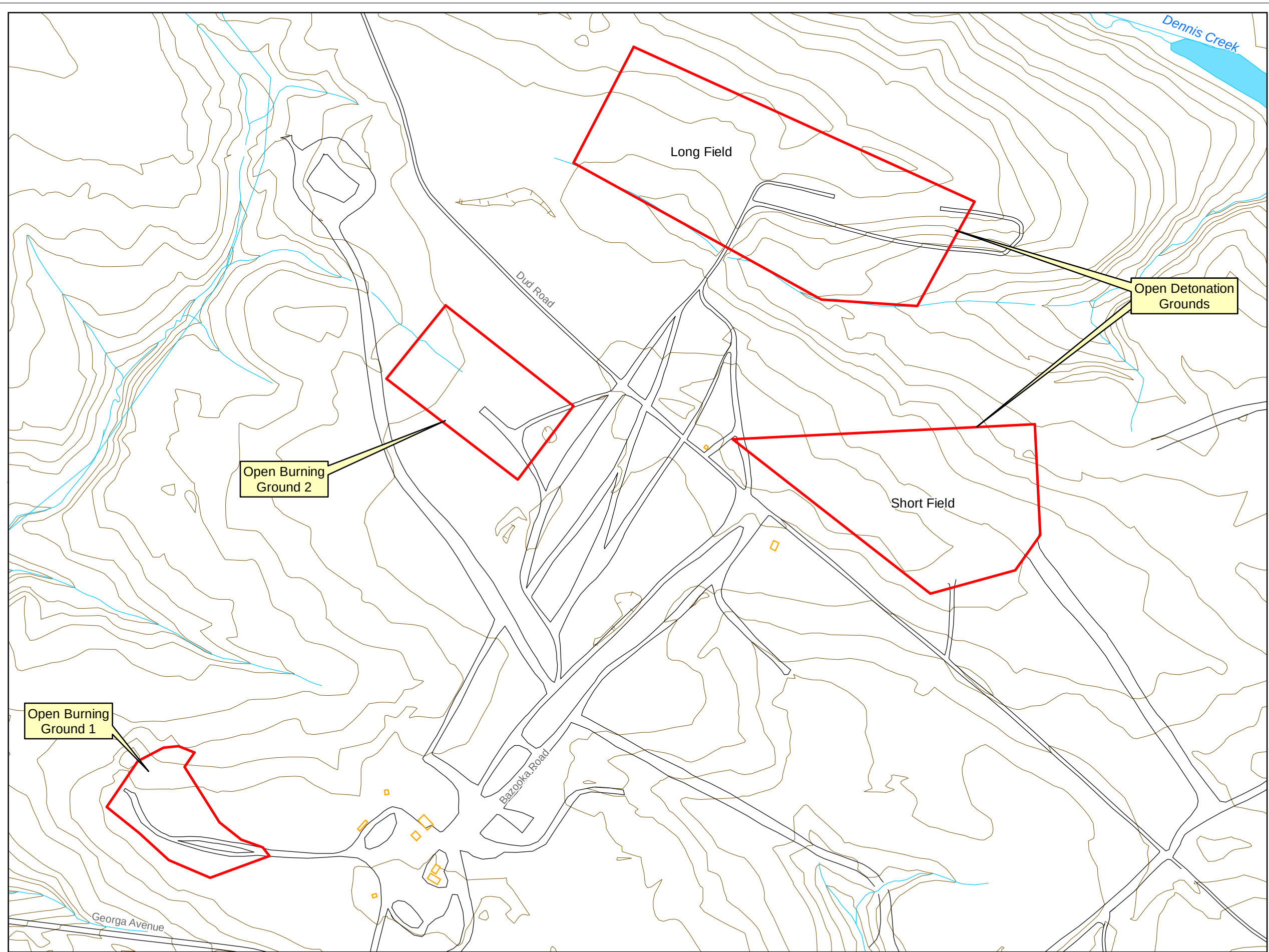
U.S. Army Corps of Engineers
Baltimore District
Letterkenny Army Depot
Chambersburg, Pennsylvania - December 2017

Figure 1 Property Location

Property location shown in blue. Base map image adapted from the U.S. Geological Survey 30x60 minute quadrangle maps, Carlisle, PA (photo inspected 1985) and Hagerstown, MD, PA, WV (1983).



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Legend

- Roads
- Creek/Streams
- Contours
- Buildings
- Open Burning/Open Detonation Areas
- Surface Water Bodies



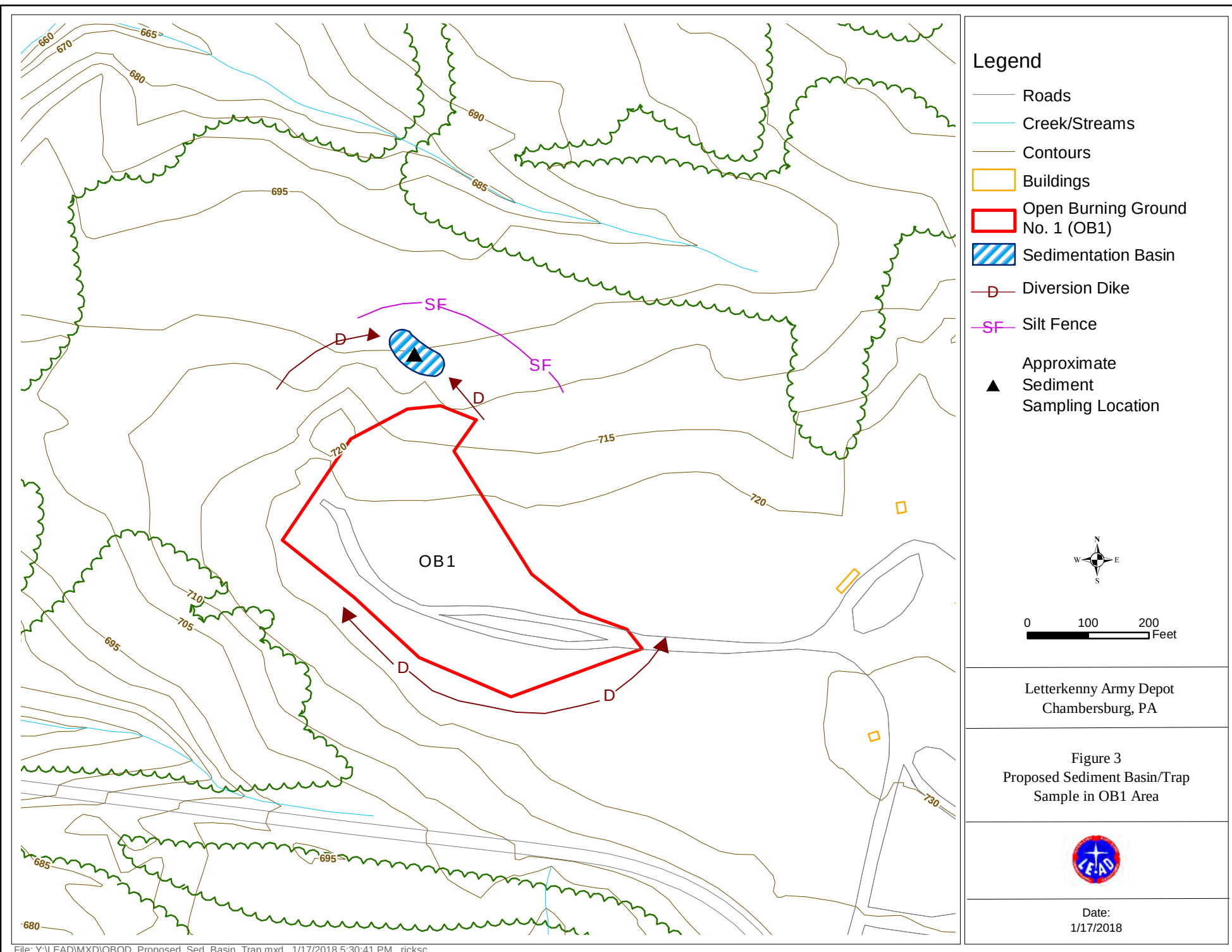
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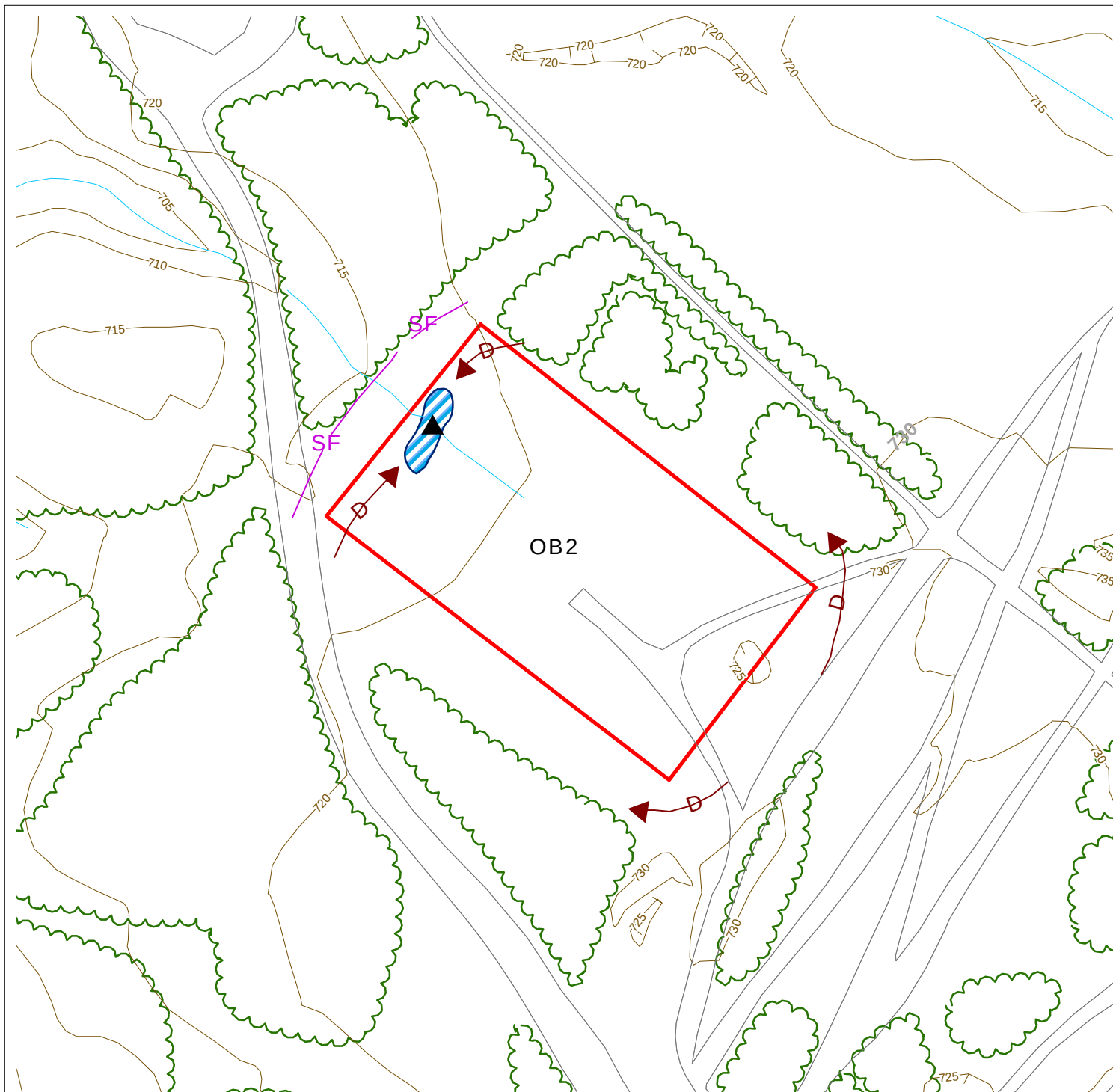
Letterkenny Army Depot
Chambersburg, PA

Figure 2
Open Burning and
Open Detonation OB/OD Areas



Date:
1/16/2018





Legend

- Roads
- Creek/Streams
- Contours
- Buildings
- Open Burning Ground No. 2 (OB2)
- ▨ Sedimentation Basin
- ▲ Approximate Sediment Sampling Location



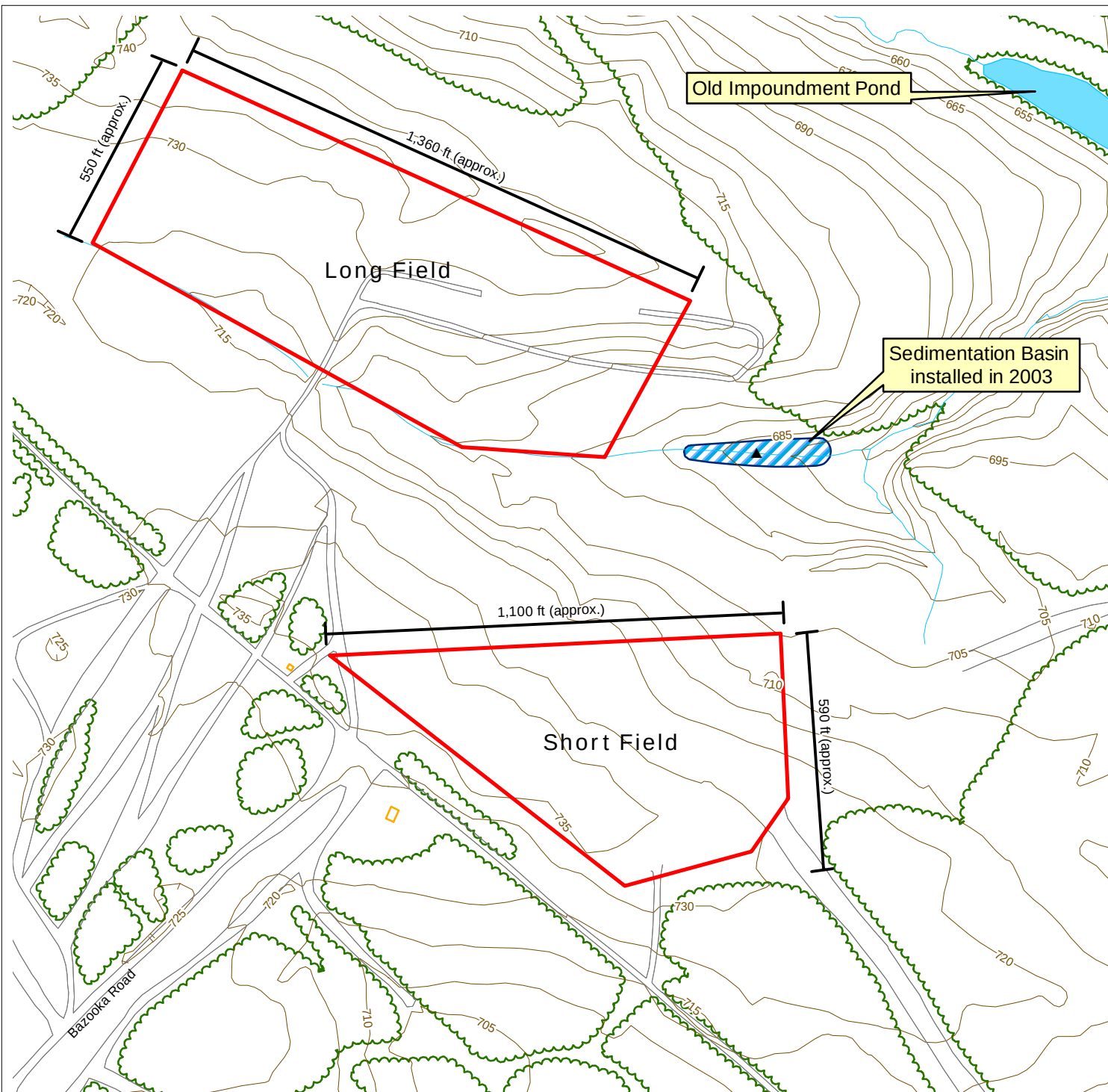
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Letterkenny Army Depot
Chambersburg, PA

Figure 4
Proposed Sediment Basin/Trap
Sample in OB2 Area



Date:
1/16/2018



Legend

- Roads
- Creek/Streams
- Contours
- Buildings
- Open Detonation Boundaries
- ▨ Sedimentation Basin
- Surface Water Bodies
- ▲ Approximate Sediment Sampling Location



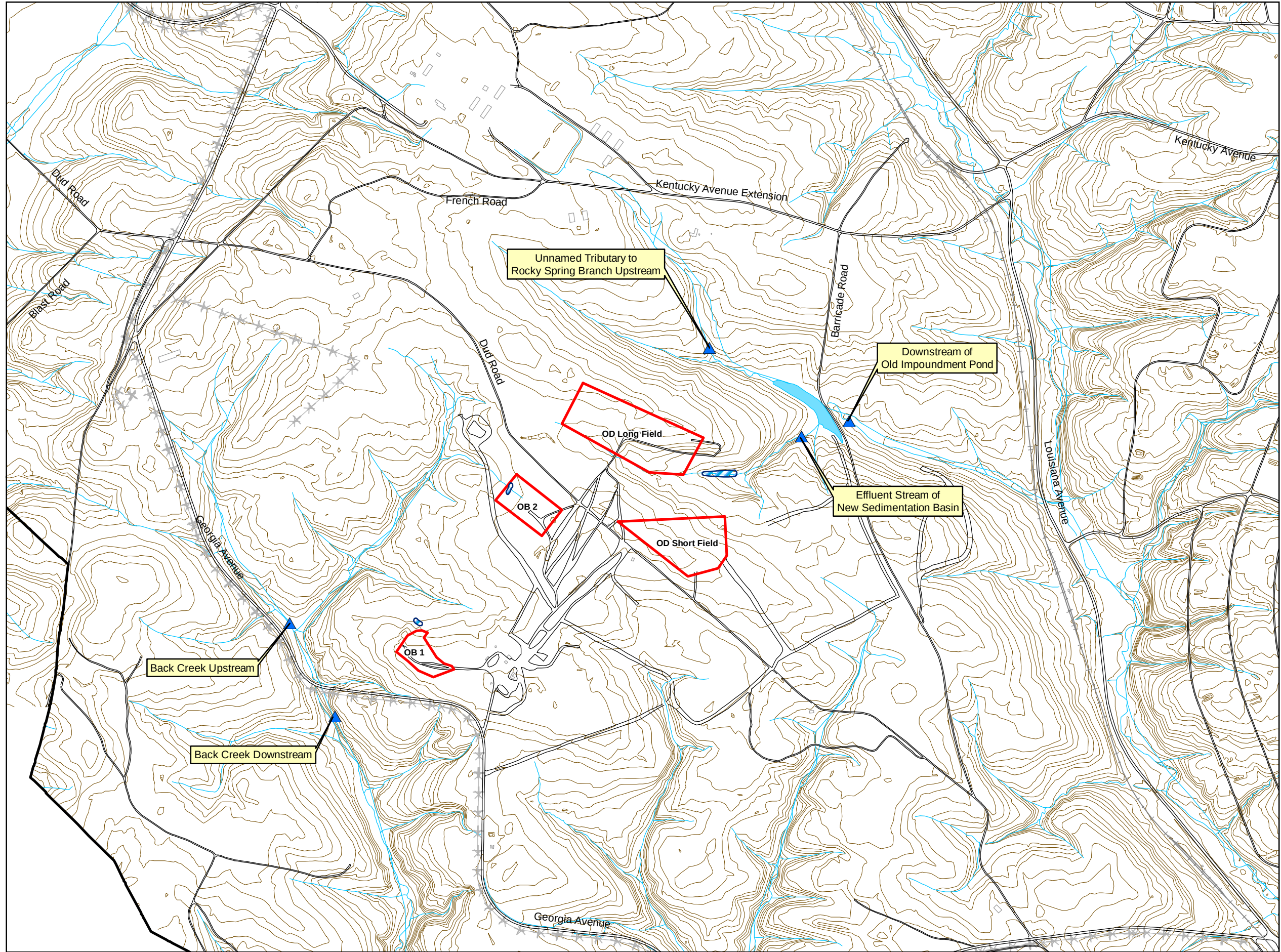
0 175 350 Feet

Letterkenny Army Depot
Chambersburg, PA

Figure 5
Proposed Sediment Basin Sample
in OD Area



Date:
1/16/2018



Legend

- Letterkenny Army Depot Boundary
- Rail
- Roads
- Fence Line
- Creek/Streams
- Contours
- Surface Water Bodies
- Open Burning/Open Detonation Areas
- Sedimentation Basin installed in 2003
- Approximate Surface Water Sampling Location



0 500 1,000 2,000 Feet

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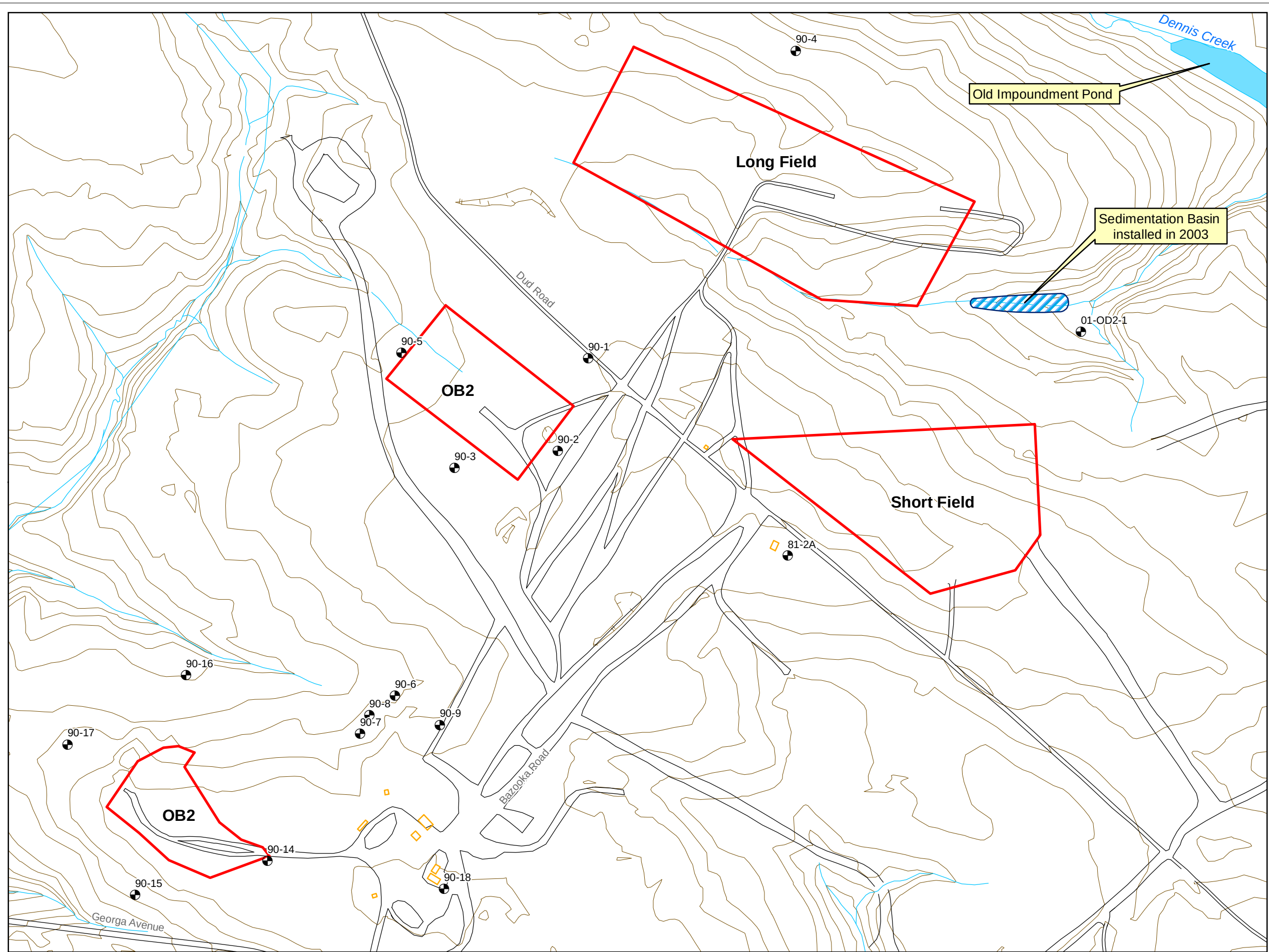
Figure 6
Proposed Stream Surface Water
Sampling Locations



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Baltimore District



Date:
1/16/2018



Legend

- Roads
- Creek/Streams
- Contours
- Buildings
- Open Burning/Open Detonation Areas
- ▨ Sedimentation Basin installed in 2003
- Surface Water Bodies
- Monitoring Well Location



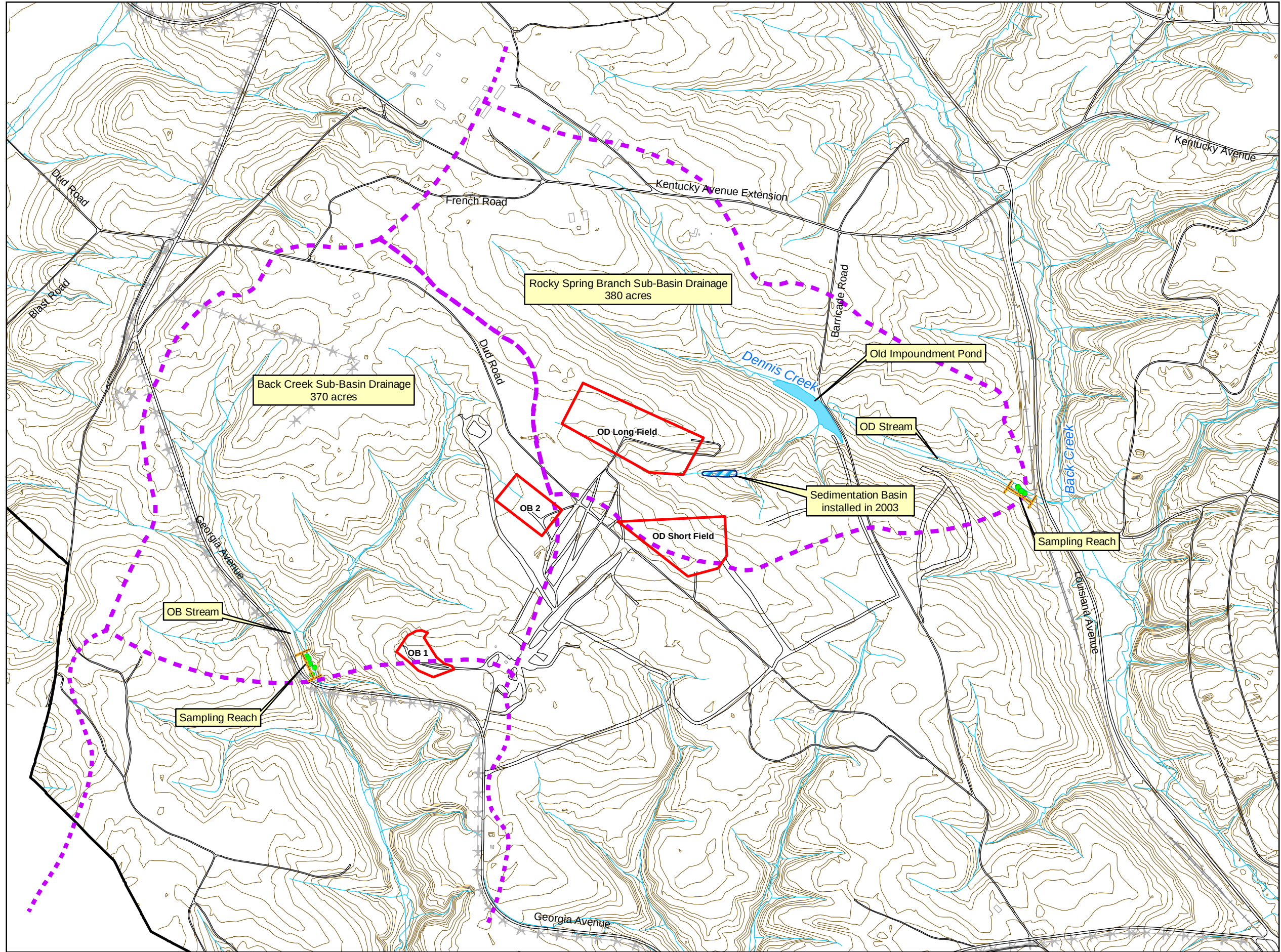
0 175 350 700 Feet

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Chambersburg, PA

Figure 7
Groundwater Monitoring
Well Locations



Date:
1/16/2018



Legend

- Letterkenny Army Depot Boundary
- Rail
- Roads
- Fence Line
- Creek/Streams
- Contours
- Surface Water Bodies
- Drainage Sub-Basin Divide
- Open Burning/Open Detonation Areas
- Sedimentation Basin installed in 2003
- Benthic Macroinvertebrate Surber Sample (2002)
- Benthic Macroinvertebrate Sample Reach



0 500 1,000 2,000 Feet

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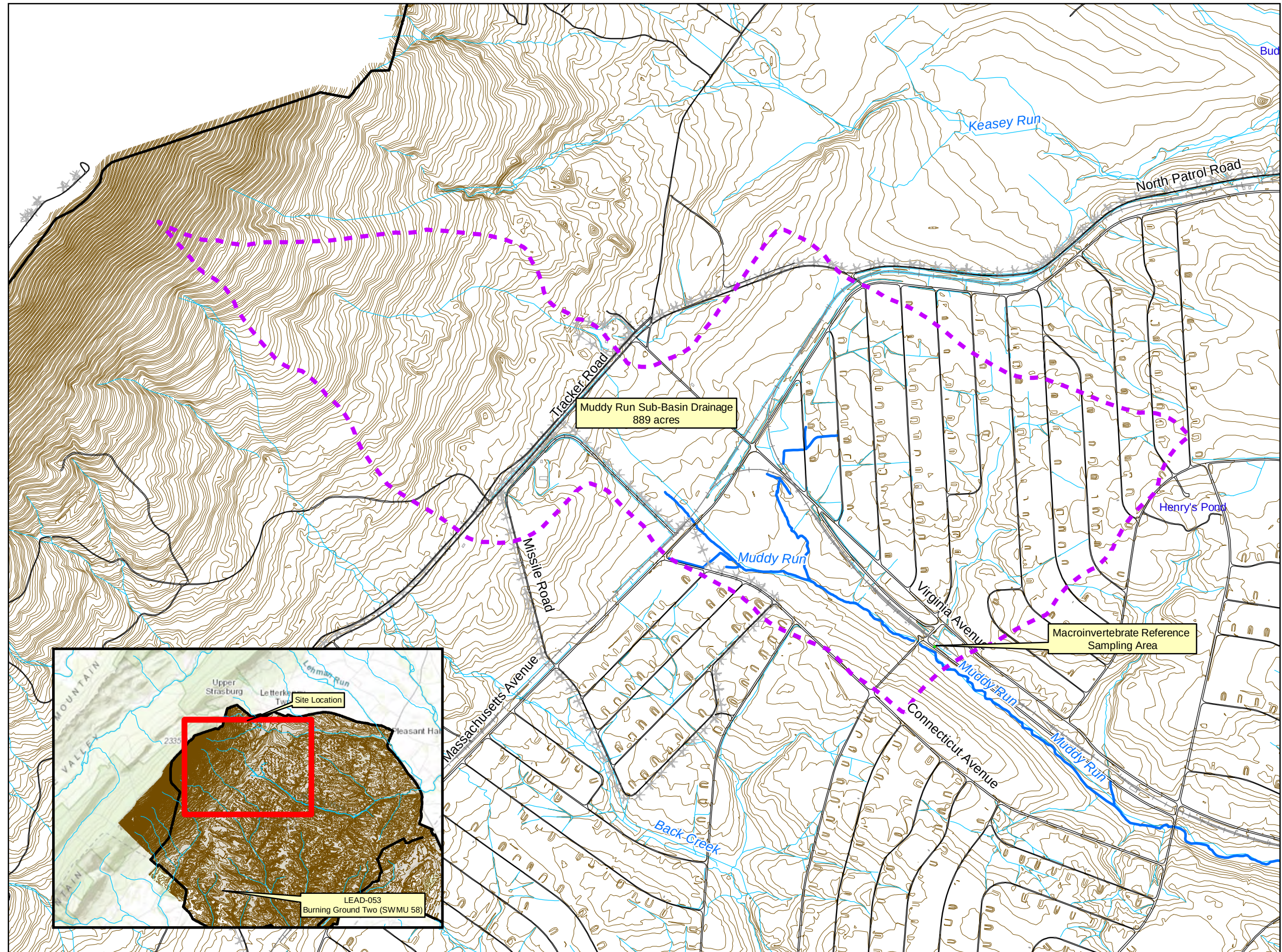
Figure 8
Benthic Macroinvertebrate
Compliance Sampling Locations



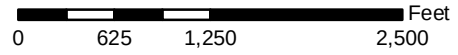
U.S. Army Corps of Engineers
Baltimore District



Date:
1/16/2018



- Legend**
- Letterkenny Army Depot Boundary
 - Rail
 - Roads
 - Fence Line
 - Muddy Run
 - Creek/Streams
 - Contours
 - Drainage Sub-Basin Divide



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Figure 9
Muddy Run Benthic Macroinvertebrate
Reference Sampling Location



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Date:
1/16/2018